

Mathematical Analysis 2

Seminar 7

Improper integrals (2)

Abel-Dirichlet convergence test:

$$-\infty < a < b \leq \infty$$

$f, g : [a, b) \rightarrow \mathbb{R}$ locally integrable such that:

$$(A) \begin{cases} f \text{ is continuous, with a bounded primitive} \\ g \text{ is decreasing, } \lim_{x \rightarrow b} g(x) = 0 \end{cases}$$

or

$$(B) \begin{cases} \int_a^b f(x) dx \text{ is convergent} \\ g \text{ is decreasing, } \lim_{x \rightarrow b} g(x) \in \mathbb{R} \end{cases}$$

Then $\int_a^b f(x)g(x) dx$ is convergent.

Example 1 $\int_0^\infty \sin x e^x dx$

$$e^x = t; x = \ln t; dx = \frac{1}{t} dt$$

$$I = \int_1^\infty \frac{\sin t}{t} dt$$

$$f(t) = \sin t \text{ continuous; } \int \sin t dt = -\cos t$$

$$g(t) = \frac{1}{t} \searrow; \lim_{t \rightarrow \infty} g(t) = 0$$

$$\Rightarrow \int_1^\infty \frac{\sin t}{t} dt \text{ is convergent. } \uparrow \text{ bounded}$$

Remark. The Abel-Dirichlet convergence test can be formulated for:
 $-\infty \leq a < b < \infty$ with $[a, b]$ replaced by $(a, b]$ and $\lim_{x \rightarrow b}$ replaced by $\lim_{x \rightarrow a}$.

Problem 1. Study the convergence of $\int_0^\infty \frac{\sin x}{x^p} dx$ ($p > 0$).

$$I = \underbrace{\int_0^1 \frac{\sin x}{x^p} dx}_{I_1} + \underbrace{\int_1^\infty \frac{\sin x}{x^p} dx}_{I_2}$$

$I_1: \lim_{x \rightarrow 0} x^\alpha \cdot \frac{\sin x}{x^p} \stackrel{\alpha=p-1}{=} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

(C2) I_1 is convergent $\Leftrightarrow \alpha < 1 \Leftrightarrow p-1 < 1 \Leftrightarrow p < 2$

$I_2: f(x) = \sin x$ continuous on $[1, \infty)$

$\int \sin x dx = -\cos x$ is bounded ($\cos x \in [-1, 1], \forall x$) $\left(\begin{array}{l} (A-D) \\ \Rightarrow I_2 \text{ is convergent} \\ \forall p > 0 \end{array} \right.$

$g(x) = \frac{1}{x^p} \searrow; \lim_{x \rightarrow \infty} g(x) = 0$ ($p > 0$)

I is conv. $\Leftrightarrow I_1, I_2$ are conv. $\Leftrightarrow p \in (0, 2)$

Problem 2. Study the convergence of $\int_0^\infty \frac{\cos x}{x^p} dx$ ($p > 0$).

$$\int_0^\infty \dots = \underbrace{\int_0^1 \dots}_{I_1} + \underbrace{\int_1^\infty \dots}_{I_2}$$

I_2 is convergent, $\forall p > 0$ (just like in Problem 1)

$I_1: \lim_{x \rightarrow 0} x^\alpha \cdot \frac{\cos x}{x^p} \stackrel{\alpha=p}{=} \lim_{x \rightarrow 0} \cos x = 1$

(C2) $L=1$; I_1 conv. $\Leftrightarrow \alpha < 1 \Leftrightarrow p < 1$

Convergence $\Leftrightarrow p \in (0, 1)$

Problem 3. Compute $\int_0^\infty \frac{\ln x}{x^2+1} dx$.

$\ln \frac{1}{x} = -\ln x$
 $x = \frac{1}{t}; dx = -\frac{1}{t^2} dt \quad (t = \frac{1}{x})$
 $I = \int_0^\infty \frac{\ln \frac{1}{t}}{\frac{1}{t^2}+1} \cdot \left(-\frac{1}{t^2}\right) dt = \int_0^\infty \frac{-\ln t}{1+t^2} dt = -I \Rightarrow 2I = 0 \Rightarrow I = 0$
 $\uparrow$ I is convergent. (independent study)

$\textcircled{I} = \int_{-a}^a f(x) dx \stackrel{x=-t}{=} \int_a^{-a} f(-t) (-dt) =$
 $\uparrow f \text{ is odd}$
 $= \int_a^{-a} -f(t) dt = -\textcircled{I} \Rightarrow \boxed{I=0}$

Problem 4. Compute $\int_0^\infty \frac{\ln x}{x^2+4} dx$.

$\ln \frac{1}{x} = -\ln x$
 $x = \frac{1}{t}; dx = -\frac{1}{t^2} dt \quad (t = \frac{1}{x})$
 $I = \int_0^\infty \frac{\ln \frac{1}{t}}{\frac{1}{t^2}+4} \cdot \left(-\frac{1}{t^2}\right) dt = \int_0^\infty \frac{-\ln t}{1+4t^2} dt = -I \Rightarrow 2I = 0 \Rightarrow I = 0$
 $\uparrow$ I is convergent. (independent study)

The previous method does not work

$x^2+4 = 4\left(\frac{x^2}{4}+1\right); \frac{x}{2} = t$
 $= 4(t^2+1)$
 $I = \int_0^\infty \frac{\ln 2t}{4t^2+4} 2 dt = \int_0^\infty \frac{\ln 2 + \ln t}{2(t^2+1)} dt$
 $= \frac{1}{2} \left(\int_0^\infty \frac{\ln 2}{t^2+1} dt + \int_0^\infty \frac{\ln t}{t^2+1} dt \right) = \frac{\ln 2}{2} \int_0^\infty \frac{1}{t^2+1} dt = \frac{\ln 2}{2} \arctan t \Big|_0^\infty = \frac{\pi \ln 2}{4}$

Problem 5. Compute $\int_0^\infty \frac{\arctan x}{x^2+x+1} dx$. $\arctan \frac{1}{x} = \frac{\pi}{2} - \arctan x \quad (x > 0)$

$x = \frac{1}{t} \quad (t = \frac{1}{x}); dx = -\frac{1}{t^2} dt \Rightarrow I = \int_0^\infty \frac{\arctan \frac{1}{t}}{\frac{1}{t^2} + \frac{1}{t} + 1} \cdot \left(-\frac{1}{t^2}\right) dt =$
 $= \int_0^\infty \frac{\frac{\pi}{2} - \arctan t}{1+t+\frac{1}{t}} dt$
 $I = \frac{\pi}{2} \int_0^\infty \frac{1}{t^2+t+1} dt - I \Rightarrow 2I = \frac{\pi}{2} \int_0^\infty \frac{1}{t^2+t+1} dt$
 $\uparrow$ I convergent. (independent study)

$I = \frac{\pi}{4} \int_0^\infty \frac{1}{\left(t+\frac{1}{2}\right)^2 + \frac{3}{4}} dt = \frac{\pi}{4} \int_{1/2}^\infty \frac{1}{u^2 + \left(\frac{\sqrt{3}}{2}\right)^2} du = \frac{\pi}{4} \cdot \frac{2}{\sqrt{3}} \arctan \frac{2u}{\sqrt{3}} \Big|_{1/2}^\infty$
 $= \frac{\pi}{2\sqrt{3}} \left(\frac{\pi}{2} - \arctan \frac{1}{\sqrt{3}} \right) = \frac{\pi}{2\sqrt{3}} \left(\frac{\pi}{2} - \frac{\pi}{6} \right) = \frac{\pi^2}{6\sqrt{3}}$

$\boxed{I = \frac{\pi^2}{6\sqrt{3}}}$

Problem 6. Compute $\int_0^{\pi/2} \ln(\cos x) dx$ and $\int_0^{\pi/2} \ln(\sin x) dx$. *Prove that I_2 is convergent!*

$I_1: x = \frac{\pi}{2} - t; dx = -dt; I_1 = \int_{\pi/2}^0 \ln\left(\cos\left(\frac{\pi}{2}-t\right)\right) (-dt) = I_2 \quad (I_1 = I_2 = I)$
 $I_1 + I_2 = \int_0^{\pi/2} \ln(\sin x \cdot \cos x) dx = \int_0^{\pi/2} \ln\left(\frac{\sin 2x}{2}\right) dx = \int_0^{\pi/2} \ln(\sin 2x) dx - \int_0^{\pi/2} \ln 2 dx$
 $2I = \int_0^{\pi/2} \ln(\sin t) \cdot \frac{1}{2} dt - \ln 2 \cdot \frac{\pi}{2} = \frac{1}{2} \left(\int_0^{\pi/2} \ln(\sin t) dt + \int_{\pi/2}^\pi \ln(\sin t) dt \right) - \frac{\pi \ln 2}{2}$
 $= \frac{1}{2} I + \frac{1}{2} \int_0^{\pi/2} \ln\left(\frac{\sin(\frac{\pi}{2}+x)}{\cos x}\right) dx - \frac{\pi \ln 2}{2}$
 $2I = \frac{1}{2} I + \frac{1}{2} I - \frac{\pi \ln 2}{2} \Rightarrow \boxed{I = -\frac{\pi \ln 2}{2}}$

Problem 7. Compute $\int_{\textcircled{0}}^1 \frac{\ln(1+x)}{x} dx$.

$$\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1 \quad (\Rightarrow \text{conv.})$$

$$\begin{aligned} \frac{\ln(1+x)}{x} &= \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} x^{n-1}, \quad x \in (-1, 1] \quad \left| \int_0^1 dx \right. \\ I &= \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \int_0^1 x^{n-1} dx = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \cdot \frac{x^n}{n} \Big|_0^1 \quad \cancel{dx} = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n^2} \\ &= 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}. \quad (\text{MA1, semester 1, Fourier series}). \end{aligned}$$